The Systems Behind Modern AI

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Abstract

Rendering computer graphics requires a large volume of floating point operations (FLOPs) to be performed in parallel. This task is not feasible for standard CPUs. GPU's were purpose built in the late 20th century as a solution to meet these demands. Initially, the design of GPU's was justifiably rigid, baking graphics operations in as hardware primitives. Despite this, the demand for scientific compute led researchers and developers to exploit graphics primitives for general purpose parallel processing. Thankfully, in 2006, the increasing complexity of graphics pipelines and demand for large scale numerical linear algebra led NVIDIA to entirely redesign